

COLLAPSED PRIMAL-DUAL APPROXIMATION OF LARGE BINARY INTEGER LINEAR PROGRAMS

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We have been using the collapsed primal-dual approach to approximately solve large marketing optimization (generalized assignment) problems since 2004, including marketing campaigns to millions of customers at Discover Financial Services in the late 2000s. Our motivation was the high cost and limitations of commercial solvers. More recently, we extended this methodology to constrained k-means clustering. Here we make a natural extension to large binary integer linear programs (BILPs), including significant applications like the 0-1 multidimensional knapsack problem. **Williamson** provides an overview of the general primal-dual approach:

- Formulate the primal binary integer linear program
- Relax the primary BILP to a primary linear program with 0-1 constraints replaced by interval $[0,1]$
- Formulate and solve the dual of the relaxed primal LP
- Use the dual LP solution together with complementary slackness to approximate the BILP solution
- Dual LP objective function provides a bound to the primal BILP objective function, thereby providing an approximation ratio measuring BILP solution precision

We follow Williamson's general recipe except that we also approximately solve the dual in collapsed form, thereby avoiding a large-scale LP solver for large BILP problems. **Li, Sun, and Ye** have recently advocated the collapsed dual approach, but they seem to conclude that a special algorithm is required. Actually, the collapsed dual can be solved via a simple, plain-vanilla nonlinear minimizer and the BILP can then be approximated with good large-problem precision, in many cases, via a simple heuristic sort using dual shadow prices associated with binary constraints. Indeed, the simplicity and precision of this approach for large, unwieldy problems is its primary attraction. This approach has limitations, however, particularly for pure set covering problems.

The standard maximization BILP, which includes the 0-1 multidimensional knapsack problem, is:

$$\text{Maximize: } \sum_{j=1}^n c_j x_j$$

$$\text{Subject to: } \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m$$

$$x_j \in \{0, 1\}, \quad j = 1, \dots, n$$

where normally $n \gg m$ and we assume that $c_j > 0$, $b_i > 0$, and $a_{ij} \geq 0$ so that the initial solution with all $x_j = 0$ is feasible. This BILP can be relaxed to a linear program by replacing the binary constraints with $0 \leq x_j \leq 1$, $j = 1, \dots, n$. The dual of this relaxed LP is

$$\text{Minimize: } \sum_{i=1}^m b_i y_i + \sum_{j=1}^n z_j$$

$$\text{Subject to: } \sum_{i=1}^m a_{ij} y_i + z_j \geq c_j, \quad j = 1, \dots, n$$

$$y_i, z_j \geq 0, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

but clearly at a minimum $z_j = \max\left(0, c_j - \sum_{i=1}^m a_{ij} y_i\right)$ leading to the collapsed dual

$$\text{Minimize: } \sum_{i=1}^m b_i y_i + \sum_{j=1}^n \max\left(0, c_j - \sum_{i=1}^m a_{ij} y_i\right)$$

$$\text{Subject to: } y_i \geq 0, \quad i = 1, \dots, m$$

This objective function is not everywhere smooth, but it is continuous and convex, and it is solvable via a simple, plain-vanilla nonlinear minimizer like *nlinb* in R. It is even possible to formulate a usable

gradient function. If y^* denotes the quasi-optimal dual solution, we then rank-order $c_j - \sum_{i=1}^m a_{ij} y_i^*$ descending and make binary $x_j = 1$ allocations until they are no longer feasible. For large BILPs, approximations are quasi-optimal, as guaranteed by the dual upper bound $\sum_{i=1}^m b_i y_i^* +$

$\sum_{j=1}^n \max\left(0, c_j - \sum_{i=1}^m a_{ij} y_i^*\right)$. We were able to approximate a large BILP with 300 constraints and

50,000 variables in about 57 minutes on a beefy Dell XPS Tower PC. The resulting BILP value was within 99.9% of the dual upper bound. We provide a simple R-function `BILP.MAX.APPROX` for this task. For comparison, we also provide R-functions `BILP.MAX.EXACT` and `BILP.MAX.LP` for problems where the `R lp` function in the `lpSolve` package can solve the BILP exactly or else solve the relaxed LP explicitly for a refined approximation, although in most cases explicit solution of the relaxed LP doesn't add much or any precision to the BILP approximation.

Now consider the mirror-image BILP minimization problem which includes various minimum-cost covering problems. Once again, we assume that $c_j > 0$, $b_i > 0$, and $a_{ij} \geq 0$.

$$\text{Minimize: } \sum_{j=1}^n c_j x_j$$

$$\text{Subject to: } \sum_{j=1}^n a_{ij} x_j \geq b_i, \quad i = 1, \dots, m$$

$$x_j \in \{0, 1\}, \quad j = 1, \dots, n$$

Putting $x_j = 1 - t_j$, we get the equivalent problem

$$\text{Maximize: } \sum_{j=1}^n c_j t_j$$

$$\text{Subject to: } \sum_{j=1}^n a_{ij} t_j \leq \sum_{j=1}^n a_{ij} - b_i, \quad i = 1, \dots, m$$

$$t_j \in \{0, 1\}, \quad j = 1, \dots, n$$

where we assume that an n -vector of ones is feasible for the minimization problem. If $\{t_j^*\}$ is a

quasi-optimal solution to the maximization problem, then $\{x_j^*\} = \{1 - t_j^*\}$ is a quasi-optimal

solution to the mirror-image BILP minimization problem. We provide analogous R-functions

`BILP.MIN.EXACT`, `BILP.MIN.LP`, and `BILP.MIN.APPROX` which incorporate their corresponding maximizers.

References

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The following references relate to example problems pulled from J. E. Beasley's OR-Library people.brunel.ac.uk/~mastijb/jeb/info.html

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The following references relate to other applications of the collapsed primal-dual approach.

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github.com/raagnew/Marketing-Optimization-in-Python

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